

Time Series Forecasting Project Exposé

Group ID: 16

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1 Additional Dataset

As an additional dataset, we plan to use the Electricity Load Diagrams dataset [3]. It contains electricity consumption time series for several hundred clients, making it structurally similar to the benchmark task where multiple related time series have to be forecasted. The forecasting use case is to predict future electricity demand from historical consumption patterns.

2 Model Architecture and Family

We plan to focus on lightweight neural forecasting models for multivariate time series with known future covariates.

As a first neural baseline, we intend to use DLinear [4], since it provides a simple but competitive reference model for long-horizon forecasting. Building on this, our main direction will be a Temporal Convolutional Network (TCN) [1], potentially extended with future covariates and learned series embeddings to account for unit-specific behavior.

If the TCN does not perform sufficiently well, we will consider a TSMixer-style architecture [2] as an alternative.

3 Key Concerns/Open Questions

A main concern is whether the neural models can consistently improve over strong seasonal baselines, since the target may contain stable daily and weekly patterns. Another open question is how much the known future covariates improve forecasting performance compared to using only historical target values and calendar features.

We also need to decide on suitable input window lengths, normalization strategies, and whether to predict the full horizon directly or use an iterative rollout.

References

- [1] Shaojie Bai, J. Zico Kolter, and Vladlen Koltun. An empirical evaluation of generic convolutional and recurrent networks for sequence modeling. *arXiv preprint arXiv:1803.01271*, 2018.
- [2] Si-An Chen, Chun-Liang Li, Nate Yoder, Sercan O. Arik, and Tomas Pfister. Tsmixer: An all-mlp architecture for time series forecasting. *Transactions on Machine Learning Research*, 2023.

- [3] Artur Trindade. Electricityloadaddiagrams20112014, 2015.
- [4] Ailing Zeng, Muxi Chen, Lei Zhang, and Qiang Xu. Are transformers effective for time series forecasting? *Proceedings of the AAAI Conference on Artificial Intelligence*, 37(9):11121–11128, 2023.